FedAdapt: Unsupervised Federated Domain Adaptation for Cross-Satellite Hyperspectral Image Classification

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***Abstract*—Multisatellite hyperspectral image (HSI) classification is increasingly expected to be performed on-orbit to reduce downlink and annotation costs. However, existing unsupervised domain adaptation (UDA) methods typically rely on the closed-set assumption that source and target domains share the same label space, which is often violated in real multisatellite deployments due to temporal land-cover changes and sensor upgrades. When the target domain contains previously unseen classes, directly aligning the two domains may induce negative transfer, making it difficult to learn a reliable decision boundary between known and unknown categories, especially under privacy constraints that prohibit raw data sharing. To address these challenges, we propose FedAdapt, an open-set unsupervised federated domain adaptation framework for cross-satellite HSI classification. FedAdapt first constructs class prototype vectors (CPVs) to induce a discriminative metric space for known classes and estimates an instance-level reliability weight for each target sample based on its similarity to the prototypes. Based on this reliability, a relevance-driven weighting strategy is introduced to perform cautious adversarial domain alignment, emphasizing likely known target samples while suppressing potential unknowns to mitigate negative transfer. In addition, an energy-based dynamic thresholding scheme is learned to adaptively separate known and unknown target samples without manual threshold tuning. Experimental results on multiple cross-scene HSI classification tasks demonstrate the effectiveness of our method.**

***Index Terms*—Domain adaptation, hyperspectral image (HSI) classification, multisatellite, unsupervised learning.**

I. Introduction

In RECENT years, low earth orbit (LEO) satellite constellations have been rapidly expanding [1], [2], and hyperspectral imaging has become a key capability for earth observation. Hyperspectral images (HSIs) provide rich spectral–spatial information that supports fine-grained land-cover interpretation and have been widely used in remote sensing applications [3], [4], [5]. As the number of satellites and sensing frequency grow, however, conventional "downlink-then-process" pipelines face increasing pressure [6], [7].

In practice, HSI data are typically transmitted to ground stations for storage, annotation, and model training or updating [8], [9]. This workflow incurs substantial communication overhead and relies heavily on expert labeling. For LEO missions, the contact time with a single ground station can be limited, and deploying or maintaining dense ground-station networks is expensive or even infeasible in many regions [10]. These constraints make it difficult to continuously downlink large-scale HSI data and to obtain timely, sufficient annotations [11]. As a result, there is a strong demand for on-orbit learning and adaptation, where satellites can collaboratively improve models without transferring raw data back to the ground.

Meanwhile, cross-satellite HSI classification is inevitably affected by domain shift [12], [13]. Changes in environmental conditions over time, differences in geographic scenes, and sensor upgrades in newly deployed satellites can significantly alter the data distribution, causing models trained on historical labeled data to generalize poorly to newly observed HSIs [14], [15]. Unsupervised domain adaptation (UDA) aims to mitigate such distribution mismatch by leveraging labeled data from a source domain and unlabeled data from a target domain to learn transferable representations [16], [17]. Despite encouraging progress, most UDA methods rely on the closed-set assumption that the source and target domains share the same label space [18], [19].

This assumption is frequently violated in multisatellite scenarios. Due to natural surface variability and evolving sensing conditions, the target domain may contain categories that are unseen in the labeled source domain. Formally, if

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the source label set is $\mathcal{C}_s$ and the target label set is $\mathcal{C}_t$, it is common that $\mathcal{C}_s \subset \mathcal{C}_t$ and $\mathcal{C}_t \setminus \mathcal{C}_s$ correspond to unknown classes. In this open-set setting, directly aligning the overall source and target distributions can be harmful: target samples from unknown classes may be incorrectly forced toward known-class regions, leading to negative transfer and blurred decision boundaries between known and unknown categories. Consequently, effective open-set domain adaptation (OSDA) approaches must reliably differentiate between known and unknown categories, ensuring that alignment is restricted to shared classes during the adaptation process. For HSIs, this problem is further complicated by high spectral dimensionality, strong interband correlation, and spectral similarity among different land-cover categories. Unknown target samples may therefore be close to known classes in the feature space and receive over-confident predictions from a source-trained classifier. This makes probability-based unknown detection and indiscriminate domain alignment less reliable for cross-satellite HSI classification.

Several approaches have been developed for OSDA. Saito et al. [20] introduced open-set backpropagation (OSBP), leveraging a generative model with a hard threshold, which may cause negative transfer due to a lack of domain-specific adaptation. Liu et al. [21] proposed Separate to Adapt (STA), relying solely on the source classifier, which can yield uncertain predictions for target data. Chang et al. [22] presented a mutual learning approach (MTS) that alternates between separating unknown samples and aligning distributions. Ru et al. [23] introduced OMEGA, addressing class imbalance and label shift via moving-threshold estimation and gradual alignment. Zhang et al. [24] proposed PUMCL, an OSDA framework that combines multiview consistency learning with prototypical constraints to denoise pseudo-labels and improve class discrimination across heterogeneous domains.

Most existing OSDA approaches rely on features extracted by backbone networks and utilize classifier output probabilities to identify unknown target samples [25]. This strategy presumes that known classes form compact clusters in the feature space, while unknown samples lie far from these clusters. However, this strategy can be unreliable for HSIs: unknown land-cover types may exhibit high spectral similarity to known categories, causing a source-trained classifier to make over-confident yet incorrect predictions. In addition, mainstream OSDA approaches adopt an $N+1$-dimensional classifier [24], where N is the number of source classes to jointly categorize known and unknown target samples, aggregating all unknowns into a single class and overlooking their inherent diversity and complexity. As a result, when unknown samples share feature similarities with known classes, the model struggles to learn sufficiently discriminative decision boundaries.

Beyond open-set recognition, multisatellite learning also raises privacy and operational constraints. Newly acquired HSIs may be sensitive, and transmitting raw data between satellites can be restricted by security and policy [26]. Federated learning provides a promising paradigm for collaborative training by aggregating model updates from distributed clients without requiring raw-data sharing [27]. Nevertheless, achieving collaborative UDA across multiple satellites while preserving the privacy of each satellite's data remains under-explored: one must simultaneously handle domain shift, label-space mismatch, and the risk of negative transfer under strict data locality constraints.



Fig. 1. FedAdapt in a multisatellite network: satellites collaboratively train a global model via federated learning without sending raw hyperspectral data to ground stations or exchanging data between satellites. The framework performs open-set UDA to mitigate domain bias when new satellites observe previously unseen categories.

To address the aforementioned challenges, we propose FedAdapt, an open-set unsupervised federated domain adaptation framework tailored for cross-satellite HSI classification. As shown in Fig. 1, FedAdapt is designed for on-orbit collaborative adaptation among multiple satellites without sharing raw HSI data with ground stations or exchanging data across satellites. Specifically, a class prototype vector (CPV) is constructed to map the metric space of known source-domain samples. By measuring the similarity between a target sample and the prototypes, FedAdapt computes an instance-level reliability weight indicating how likely the sample belongs to a known class. This reliability is then used to perform relevance-driven weighted domain alignment, emphasizing alignment for likely known target samples and suppressing the influence of potential unknowns to reduce negative transfer. Finally, an energy-based dynamic weighting mechanism is designed to adaptively learn the classification boundary between known and unknown samples in the feature space. Consequently, unknown samples are effectively rejected, while known samples are correctly identified. Experimental results demonstrate that our approach substantially reduces domain bias caused by the presence of unknown categories in the target domain.

To summarize, the main innovative contributions of the proposed approach can be summarized as follows.

1) We propose FedAdapt, an open-set unsupervised federated domain adaptation framework tailored for cross-satellite HSI classification. The framework enables on-orbit collaborative model adaptation across multiple satellites without sharing raw HSI data or requiring target-domain annotations.
2) We design a CPV-induced metric space and a reliability estimation mechanism to quantify whether a target sample is likely to belong to the shared known classes.

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Building upon this, a relevance-driven sample weighting strategy for domain alignment is introduced, which emphasizes alignment for target samples that are close to known prototypes and suppresses the influence of potential unknowns, thereby reducing negative transfer.

3) We develop an energy-based dynamic thresholding scheme to learn an adaptive decision boundary between known and unknown target samples in the feature space. The learned boundary supports effective unknown rejection and improves the robustness of federated open-set adaptation across heterogeneous sensors and scenes.

The organization of this article is as follows. Section II reviews related studies on domain adaptation for HSI classification, OSDA, and federated learning for remote sensing. In Section III, the proposed method is described in detail. The experiments are shown in Section IV. Finally, conclusion is shown in Section V.

II. Related Work

A. Domain Adaptation for HSI Classification

UDA addresses domain shift by aligning the data distributions between the source and target domains. By leveraging labeled samples from the source domain together with unlabeled samples from the target domain, UDA aims to learn domain-invariant representations that enhance the model's generalization capability on the target domain. It can be broadly categorized into two categories: distribution difference-based UDA and adversarial learning-based UDA. The key idea of distribution difference-based UDA methods is to minimize domain discrepancy by selecting an appropriate statistical divergence measure and optimizing it for alignment, such as the maximum mean discrepancy [28]. Recently, Shi et al. [29] proposed FUZZLE, an efficient and straightforward approach for aligning source and target domains through domain-specific prompt tuning. A Kullback–Leibler divergence-based loss function is used, which aims to reduce distribution differences in classifying similar cross-domain data by aligning domain-specific prompts throughout the training phase. Wang et al. [30] present a square loss-based framework by explicitly maximizing both discriminability and equity. Within this framework, two innovative loss functions, class weighted squares maximization and normalized squares maximization, are introduced to jointly enhance class separability and distribution balance. Adversarial learning-based methods achieve domain invariance by framing UDA as a minimax game, where the feature extractor attempts to fool the domain discriminator, while the discriminator strives to distinguish features from different domains [31], [32]. ADAPT [31] integrates text and visual prompt tuning through an adversarial learning framework that enforces mutual adaptation between the two modalities. Yao et al. [32] proposed domain consistency adversarial learning, which uses a triplet of features to learn feature-invariant representations by distinguishing whether the feature belongs to the same domain. In hyperspectral remote sensing, adversarial domain adaptation has been widely explored for cross-scene classification under the closed-set assumption. For example, TAADA [33] designs a two-branch attention feature extractor to exploit spectral–spatial characteristics of HSIs and performs adversarial adaptation via a dual-classifier discriminator. For cross-scene wetland mapping, SSWADA [34] further introduces a weighted discriminator to mitigate negative transfer from noisy/less-transferable instances and a multiclassifier target constraint to improve target predictions.

B. Open-Set Domain Adaptation

OSDA further considers the case where the target domain contains unknown classes that are not present in the source domain. Compared with closed-set UDA, OSDA is more challenging because domain alignment and known/unknown separation must be performed simultaneously. If unknown target samples are involved in global distribution alignment, they may be incorrectly aligned with source known classes, leading to negative transfer. In recent years, several methods have been investigated to address open-set scenarios. BUS [35] effectively distinguishes the boundaries between known and unknown classes using a novel dilation-erosion-based loss in a contrastive manner. SDA [36] proposes a separate domain alignment framework for open-set cross-network node classification, in which the target domain is isolated, and neighbor centers are clustered to facilitate effective adaptation. Zhou et al. [37] propose a framework to address the detection of unknown deepfake categories in open-set scenarios. By exploiting the memorization properties of deep neural networks, the method performs adaptive clustering to generate reliable pseudo-labels for unlabeled target domain samples, which are then used to iteratively refine a source-pretrained model. Most existing OSDA methods were originally developed for general computer vision or multimedia tasks. Although they provide useful mechanisms for unknown detection and negative-transfer mitigation, they do not explicitly consider the spectral–spatial characteristics of HSI s. Recently, several studies have begun to explore open-set adaptation in the HSI domain. For example, CRLN [38] proposes a teacher–student framework that derives unknown-class supervision from prediction uncertainty and further promotes known/unknown separation during cross-domain adaptation. RDFA [39] estimates target similarity to the source domain via reconstruction discrepancy, and combines weighted adversarial feature alignment with class-level centroid alignment to improve both known-class transfer and unknown recognition. However, these HSI-oriented studies are generally conducted under centralized settings. Therefore, existing methods do not directly address the federated cross-satellite open-set adaptation problem considered in this article, where raw HSI data are locally preserved, and domain alignment must be performed under label-space mismatch and data-sharing constraints.

C. Federated Learning for Remote Sensing

Federated learning is a distributed machine learning paradigm designed to enable collaborative model training while preserving data privacy [40], [41], [42], [43]. It allows multiple edge devices or isolated data centers to jointly train deep models without sharing raw data, thereby maintaining

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data locality and improving overall model performance. From a broader perspective, this collaborative training process can be viewed as knowledge transfer across distributed clients through model interaction and iterative aggregation [44], [45], [46]. Remote sensing provides a natural fit for federated learning because data are commonly held by separate parties (e.g., different regions or agencies) or generated at the edge (e.g., onboard platforms), and centralizing imagery/labels is often impractical due to privacy, security, policy, or commercial considerations [11]. Meanwhile, the client data in remote sensing are rarely identically distributed: clients may differ in label composition and data volume, and even the same land-cover category can appear with different statistics across locations and acquisition conditions, leading to noticeable concept shift over space and time [47]. Under such conditions, naive parameter averaging can be suboptimal, which has motivated remote-sensing-tailored FL strategies. Representative remote-sensing FL studies include prototype-driven designs and privacy/efficiency enhancements. FedPM [48] summarizes each class with client-side prototypes and matches them on the server to regularize local training, alleviating distribution discrepancies for pixel-wise classification. To better cope with severe heterogeneity, Fed-PHC [49] groups clients by distribution similarity and performs cluster-wise aggregation rather than enforcing a single global model for all clients. For decentralized multimodal archives, Büyüktaş et al. [50] consider modality-specific backbones and cross-modality coordination to reduce the mismatch among clients with different sensing modalities. In addition, practical deployments also emphasize communication and privacy: PRFL [51] compresses model updates via low-rank parameter decomposition to reduce transmission load, while LDP-Fed [52] injects local differential privacy noise to mitigate membership inference threats; blockchain-based FL has also been explored to strengthen robustness against poisoning attacks [53].

III. Methodology

A. Problem Definition

In this article, we focus on how to utilize newly acquired unlabeled satellite data to perform on-orbit classification tasks without the need to transmit the data back to the ground. In open-set setting, let the source domain be denoted as $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$, containing n_s labeled samples, and the target domain as $\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}$, consisting of n_t unlabeled samples. The class sets for the source and target domains are represented by $\mathcal{C}_s$ and $\mathcal{C}_t$, respectively, where $\mathcal{C}_s \subset \mathcal{C}_t$. The subset $\mathcal{C}_{t\setminus s}$ corresponds to unknown classes present in the target domain but absent from the source domain. The data distributions of the source and target domains are denoted by p and q, respectively, with $p \neq q$.

Based on this basic background, we propose the FedAdapt network architecture, which is shown in Fig. 2. The designed multisatellite UDA framework enables collaborative knowledge transfer among multiple participants for training deep learning models, without requiring any satellite to disclose their raw data. First, data x^s and x^t from the source and target domains are passed through the feature extractor E to obtain feature representations $E(x^s)$ and $E(x^t)$. Then these features are input into the CPV weighting module and the domain discriminator D. For the source-domain features $E(x^s)$, the classification loss is computed via the CPV module, while the domain discrimination loss is evaluated by the discriminator in the server. For target domain features $E(x^t)$, the CPV module estimates an instance-level reliability weight w_t, indicating the likelihood that a sample belongs to the known classes. This reliability weight is then used to modulate the domain discrimination loss for each target sample, thereby reducing the risk of negative transfer. The three main components of FedAdapt are connected through this reliability estimation. The CPV-induced metric space defines the geometry of known land-cover classes, the prototype distance estimates the reliability of each target sample, and this reliability further guides both adversarial alignment and dynamic threshold learning. Thus, likely known target samples are adaptively aligned, while potential unknown samples are prevented from dominating the adaptation process.

B. CPV and Discriminative Loss

To enhance the discriminative power and open-set robustness of the feature space, we explicitly construct an independent CPV for each known class. Each prototype vector is initialized as an orthogonal basis in a high-dimensional space and scaled by a separation coefficient ϕ to control its magnitude. This design allows for flexible adjustment of interclass distances, resulting in compact intraclass distributions and well-separated interclass boundaries, while reserving sufficient space for the representation of unknown classes.

Formally, for K known classes, the set of CPV is defined as

$$C = \{\phi \cdot e_1, \ \ldots, \phi \cdot e_K\} \tag{1}$$

where e_k is the standard basis vector for class k, and ϕ is a tunable scaling coefficient. Here, ϕ acts as a prototype scaling factor that controls the overall separation among CPVs. A larger ϕ increases interclass margins in the CPV-induced metric space, which typically yields more compact known-class regions while reserving more open space for potential unknown samples.

To accurately measure the similarity between the extracted feature vector and each class prototype, a generalized Minkowski distance is employed as the class-wise metric. The extracted feature $E(x)$ is further processed by a fully connected layer to obtain a $|\mathcal{C}_s|$-dimensional embedding, denoted as $A(E(x))$. Let $z = A(E(x)) \in \mathbb{R}^K$ denote the projected embedding of a sample in the CPV space, and let $\{c_i\}_{i=1}^K$ represent the set of CPV corresponding to the known classes. The distance between z and the prototypes is defined as

$$\mathbf{d}(z) = \left[d_p(z, c_1), d_p(z, c_2), \ldots, d_p(z, c_K)\right]^\top \tag{2}$$

the Minkowski distance is defined as

$$d_p(z, c_j) = \|z - c_j\|_p = \left(\sum_{k=1}^{K} \left|z_k - c_{j,k}\right|^p\right)^{\frac{1}{p}} \tag{3}$$

where $c_j \in \mathbb{R}^K$ denote the jth class prototype, and p is a tunable parameter that controls the shape of the metric space.

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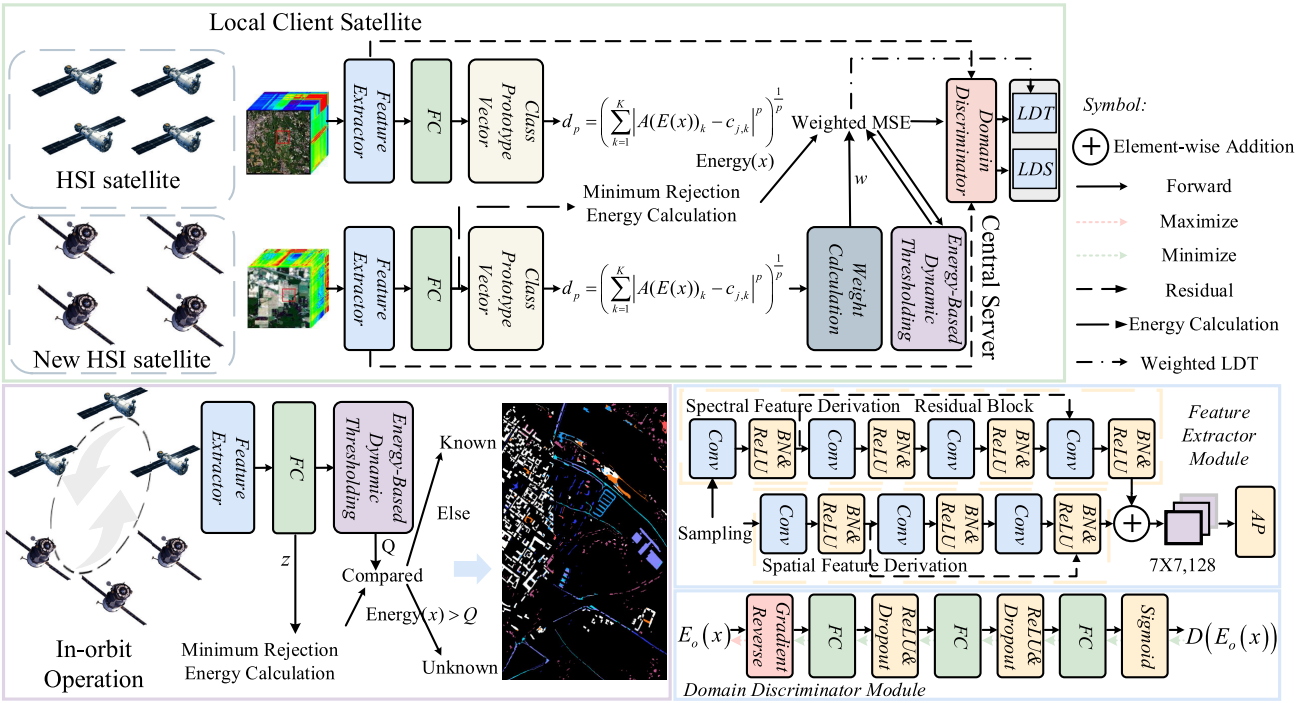


Fig. 2. Overview of the FedAdapt architecture. Client-side prototype-based learning estimates target-sample reliability, while the server conducts relevance-weighted adversarial domain alignment; an energy-based dynamic threshold is learned to separate known from unknown target samples.

The order p determines the geometry of the distance metric and thus affects the sensitivity of prototype matching, which in turn influences how target samples are weighted and aligned in the subsequent adaptation stage.

Prototype aggregation discriminative (PAD) loss is introduced to enhance feature compactness and interclass separability. For each sample, PAD loss aggregates the similarity comparisons between the sample feature and all class prototypes. The objective is to minimize the distance to its own class prototype while maximizing the distance to prototypes of all other classes. Formally, the PAD loss is defined as

$$\mathcal{L}_{\mathrm{PAD}} = \frac{1}{N}\sum_{i=1}^{N}\log\left[1+\sum_{j\neq y_i}\exp\left(d_p\left(A\left(E\left(x_i\right)\right),c_{y_i}\right)-d_p\left(A\left(E\left(x_i\right)\right),c_j\right)\right)\right] \tag{4}$$

where x_i denotes the ith sample, y_i is its ground-truth class label, c_{y_i} is the prototype of its class, and N is the sample size.

C. Relevance-Driven Sample Weighting for Domain Alignment

To quantify the reliability of each target-domain sample with respect to the known classes, reciprocal weighting normalization is developed for weight assignment. For each target sample, let l denote the minimum distance to all class prototypes, formulated as $l = \min_j d_p(A(E(x^t)), c_j)$. The reliability weight w_t is defined as follows:

$$\tilde{w}_i = \frac{1}{l_i+\epsilon} \tag{5}$$

$$w_t = \frac{\tilde{w}_i - \min(\tilde{w})}{\max(\tilde{w}) - \min(\tilde{w}) + \epsilon} \tag{6}$$

where $l_i = \min_j d_p(A(E(x_i^t)), c_j)$ and the min/max in (6) are taken over $\{\tilde{w}_i\}$ within the current optimization step. ϵ is a small constant for numerical stability.

The domain discriminator $D(\cdot)$ is trained to distinguish between domains using a binary cross-entropy objective. The adversarial losses for the source and target domains are formulated as

$$\mathcal{L}_{\mathrm{adv}}^{s} = \frac{1}{M_s}\sum_{i=1}^{M_s} L_{\mathrm{bce}}\left(D\left(E_{o_s}\left(x_i^s\right)\right), d_i^s\right) \tag{7}$$

$$\mathcal{L}_{\mathrm{adv}}^{t} = \frac{1}{M_t}\sum_{j=1}^{M_t} w_t^j L_{\mathrm{bce}}\left(D\left(E_{o_t}\left(x_j^t\right)\right), d_j^t\right) \tag{8}$$

where d_i^s and d_j^t are the domain labels for source and target samples, respectively. M_s is the number of source-domain samples, and M_t is the number of target-domain samples. The overall domain adversarial loss is thus given by

$$\mathcal{L}_{\mathrm{DA}} = \mathcal{L}_{\mathrm{adv}}^{s} + \mathcal{L}_{\mathrm{adv}}^{t}. \tag{9}$$

This relevance-driven scheme ensures that target samples with higher affinity to known classes exert greater influence during domain alignment, while samples with low relevance (potential outliers or unknowns) contribute less, thereby reducing the risk of negative transfer and improving federated open-set robustness in UDA. This design is different from existing weighted adversarial alignment methods. IWAN [54] mainly reweights source samples in partial DA to suppress source outlier classes, whereas FedAdapt focuses on target samples and aims to reduce the influence of potential unknown

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samples in the open set target domain. Compared with STA [21] and DAMC [55], which obtain target weights from separation procedures or classifier-based similarity, FedAdapt directly computes the weight from the minimum distance to the CPV induced known class space and uses it only in the target adversarial loss. Different from UADAL [56], FedAdapt does not introduce a three-way discriminator, but keeps a standard discriminator and combines prototype-based reliability weighting with the energy-based dynamic threshold.

D. Energy-Based Dynamic Thresholding Learning

Fixed thresholds for distinguishing known from unknown classes are highly sensitive to data characteristics and are difficult to tune for robust performance. We therefore present an energy-based dynamic thresholding method that adaptively models the decision boundary of known classes in the target domain during training. This mechanism relies on a trainable parameter Q, which represents the boundary of known-class features. A sample is rejected as unknown when its rejection score exceeds Q.

Specifically, for a target-domain sample (x^t), we first obtain its projected feature vector ($z = A(E(x^t)) \in \mathbb{R}^K$). The energy score is then defined as:

$$Energy\left(x^t\right) = -\log\left(\sum_{k=1}^{K}\exp\left(z_k\right)\right). \tag{10}$$

Lower energy value indicates higher confidence in belonging to a known class, while a higher value suggests uncertainty.

During training, we encourage the threshold to separate high-confidence samples from low-confidence samples, using a weighted mean squared error loss. The reliability weight w^t is used to reduce the influence of unknown samples. The loss function is given by

$$\mathcal{L}_Q^t = \frac{1}{N_t}\sum_{i=1}^{N_t} w_i^t\left(Q - \mathrm{sg}\left[Energy\left(x_i^t\right)\right] - g\right)^2 \tag{11}$$

where N_t is the number of target-domain samples in a training step, g denotes a hyperparameter controlling the margin between Q and $Energy(x_i^t)$, and sg[·] denotes the stop-gradient operation.

This energy-based dynamic thresholding mechanism enables the model to automatically adapt the decision boundary between known and unknown classes during training, without relying on a fixed or manually tuned threshold.

E. Training and Testing Processes

Let $\mathcal{S}$ denote the set of source satellites with labeled data and $\mathcal{T}$ denote the set of target satellites with unlabeled data. Each satellite o maintains a local feature extractor E_o and a prototype-mapping head $A_o : \mathbb{R}^d \to \mathbb{R}^K$ that projects features into a fixed CPV space $C = \{\phi e_k\}_{k=1}^K$. Each target satellite $o_t \in \mathcal{T}$ further holds a learnable rejection threshold Q_{o_t}. A central server maintains a domain discriminator D and performs federated aggregation of $\{E_o, A_o\}$; raw hyperspectral data never leave their satellites.

Algorithm 1 Training of the Proposed Method

Require: Source client set $\mathcal{S}$ with labeled datasets $\{\mathcal{D}_{o_s}\}_{o_s\in\mathcal{S}}$; target client set $\mathcal{T}$ with unlabeled datasets $\{\mathcal{D}_{o_t}\}_{o_t\in\mathcal{T}}$; hyperparameters a, b.

Ensure: Trained $\{E_o, A_o\}_{o\in\mathcal{S}\cup\mathcal{T}}$, $\{Q_{o_t}\}_{o_t\in\mathcal{T}}$, and D.

Initialize $\{E_{o_s}, A_{o_s}\}$ for all $o_s \in \mathcal{S}$ and server discriminator D.
Stage I: Source-only pretraining
for $e = 1$ **to** $Epoch_{pre}$ **do**
  for all $o_s \in \mathcal{S}$ **in parallel do**
    for $\{(x_i, y_i)\} \subset \mathcal{D}_{o_s}$ **do**
      $f_i \leftarrow E_{o_s}(x_i)$; $z_i \leftarrow A_{o_s}(f_i)$
      Compute $d_p(z_i, c_j)$ and $\mathcal{L}_{PAD}^{(c_s)}$; update $\{E_{o_s}, A_{o_s}\}$.
    end for
  end for
  Server FedAvg: aggregate $\{E_{o_s}, A_{o_s}\}_{o_s\in\mathcal{S}}$ and broadcast
end for
Initialize each target client $o_t \in \mathcal{T}$ with global $\{E, A\}$; initialize Q_{o_t}
Stage II: UDA with domain alignment and threshold learning
for $e = 1$ **to** $Epoch_{adapt}$ **do**
  while training data not exhausted **do**
    On o_s: sample $\{(x_i^s, y_i^s)\}$; compute $\mathcal{L}_{PAD}^{(o_s)}$; update $\{E_{o_s}, A_{o_s}\}$.
    On o_t: sample $\{x_i^t\}$; $f_i^t \leftarrow E_{o_t}(x_i^t)$; $z_i^t \leftarrow A_{o_t}(f_i^t)$
    Compute $d_{i,j} = d_p(A_{o_t}(f_i^t), c_j)$, $l_i = \min_j d_{i,j}$, w_i^t, and $E(x_i^t)$
    Upload $\{f^s\}$ from o_s and $\{f^t, z^t, w^t\}$ from o_t to the server
    On server: compute $\mathcal{L}_{DA}$; update D.
    Apply GRL; send encoder gradients to o_s and o_t; update D; update $\{E_{o_s}, A_{o_s}\}$; update $\{E_{o_t}, A_{o_t}\}$; update Q_{o_t}.
  end while
  Server FedAvg over $\{E_o, A_o\}_{o\in\mathcal{S}\cup\mathcal{T}}$; aggregate $\{Q_{o_t}\}_{o_t\in\mathcal{T}}$; broadcast
end for

As shown in Algorithm 1, the training process consists of two stages. The first stage is a source-only pretraining stage that establishes a discriminative metric space for known classes and reserves open space for unknowns. The second stage jointly aligns source and target satellites via instance-level weighted adversarial learning while learning target-specific thresholds.

1) *First Stage:* On each source satellite $o_s \in \mathcal{S}$, the feature extractor E_{o_s} and prototype-mapping head A_{o_s} are trained using labeled source-domain samples to classify known classes while leaving room for unknown features. Let θ_E and θ_A denote the parameters of E and A, respectively. The objective for optimizing $(\hat{\theta}_E, \hat{\theta}_A)$ is

$$\left(\hat{\theta}_E, \hat{\theta}_A\right) = \arg\min_{\theta_E,\theta_A} \mathcal{L}_{PAD} \tag{12}$$

at the end of each communication round, the server aggregates $\{(\theta_{E_o}, \theta_{A_o})\}_{o\in\mathcal{S}}$ and broadcasts the global

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Algorithm 2 Testing of the Proposed Method

Require: Target client $o_t \in \mathcal{T}$ with $\{E_{o_t}, A_{o_t}, Q_{o_t}\}$; Target client set $\mathcal{T}$ with unlabeled datasets $\{\mathcal{D}_{o_t}\}_{o_t \in \mathcal{T}}$.
Ensure: Predictions $\{\hat{y}\}$ on o_t.

```
for each x do
    f ← E_{o_t}(x); z ← A_{o_t}(f); Energy(x) ← −log(Σ_k exp(z_k))
    if Energy(x) > Q_{o_t} then
        ŷ ← unknown
    else
        Compute d_j ← d_p(A_{o_t}(f), c_j) for j = 1, ..., K; ŷ ← arg min_j d_j
    end if
end for
```

parameters, which also initialize all target satellites for the second stage.

2) *Second Stage:* For joint adaptation, training steps are processed by pairing source satellites $o_s \in \mathcal{S}$ and target satellites $o_t \in \mathcal{T}$. Federated aggregation is performed at the end of each epoch. The source satellite continues optimizing $\mathcal{L}_{PAD}$ to consolidate the metric geometry. The target satellite processes unlabeled training step $\{x_i^t\}$ to produce features $f_i^t = E_{o_t}(x_i^t)$, the projected feature vector $z_i^t = A_{o_t}(f_i^t)$, and prototype distances $d_{i,j} = d_p(A_{o_t}(f_i^t), c_j)$. Let $l_i = \min_j d_{i,j}$, an instance reliability w_i^t is computed by (6) via reciprocal normalization. The energy score $Energy(x^t)$ is computed by (10). The server receives source features f^s and target features f^t, together with the projected feature vector z^t and weights w^t, and the total loss in the second stage is

$$\mathcal{L}_{total} = \mathcal{L}_{PAD} - a\mathcal{L}_{DA} + b\mathcal{L}_Q^t \tag{13}$$

where a and b are hyperparameters that balance the three objectives in Stage II. Specifically, a controls the strength of the relevance-weighted adversarial domain alignment loss $\mathcal{L}_{DA}$, while b controls the contribution of the dynamic threshold learning loss $\mathcal{L}_Q^t$. In practice, these hyperparameters are selected once per task and then shared across clients, without requiring client-specific tuning during on-orbit deployment, and the optimization objectives are

$$\begin{aligned}\left(\hat{\theta}_E, \hat{\theta}_A, \hat{\theta}_Q\right) &= \arg\min_{\theta_E, \theta_A, \theta_Q} \mathcal{L}_{total} \\ \hat{\theta}_D &= \arg\max_{\theta_D} \mathcal{L}_{total}\end{aligned} \tag{14}$$

after each epoch, the server aggregates $\{(\theta_{E_o}, \theta_{A_o})\}_{o \in \mathcal{S} \cup \mathcal{T}}$ and synchronizes thresholds by averaging $\{Q_{o_t}\}_{o_t \in \mathcal{T}}$. The prototype set C remains fixed on the satellite side.

The first stage equips source satellites with a stable metric prior over known classes, yielding reliable instance weights that mitigate negative transfer when target satellites start adaptation. The second stage enables robust open-set recognition across heterogeneous sensors while preserving raw HSI data, staying local. In the testing stage, each target satellite $o_t \in \mathcal{T}$ performs server-free inference, as shown in Algorithm 2. Class assignment for target-domain samples is carried out through an energy value decision process. If $Energy(x) > Q_{o_t}$, the sample is predicted as an unknown class; otherwise assign the nearest prototype

$$\hat{y} = \begin{cases} \text{unknown}, & Energy(x) > Q \\ \arg\min\limits_j \; d_p\left(A_{o_t}\left(E_{o_t}(x)\right), c_j\right), & \text{otherwise.} \end{cases} \tag{15}$$

IV. Experimental Study

A. Description of Datasets

Before constructing the cross-domain tasks, we followed the commonly used benchmark protocol. These protocols have already undergone the necessary dataset-level preprocessing by the data providers. In this study, we did not re-process the raw digital-number images but focused on task-level preprocessing for cross-domain learning, including invalid and noisy band removal, spectral-band matching between the source and target domains, spatial patch extraction, and band-wise normalization. All compared methods were evaluated using the same input data.

In order to verify the superiority of FedAdapt in analyzing cross-domain hyperspectral data, eight datasets were used for experiments, including Pavia University (PU), Pavia Center (PC), Houston2013 (HU13), Houston2018 (HU18), Dioni, Loukia, Ziyuan1-02D (ZY), and Gaofen-5 (GF). These datasets were systematically paired to construct four OSDA classification scenarios. Specifically, the first scenario utilizes PU as the source domain and PC as the target domain, while the second scenario involves transferring knowledge from HU13 to HU18. For the third scenario, Dioni serves as the source and Loukia as the target. For the fourth scenario, ZY serves as the source and GF as the target. This diverse set of tasks enables a thorough evaluation of the method's generalization capabilities across different domains and imaging conditions. Detailed descriptions, pseudo-color images, ground-truth maps, and sample statistics of these datasets are provided in the Supplementary Material Section II.

B. Experimental Protocol

1) Evaluation Criteria: The HOS metric is adopted to assess model performance [57]. HOS is defined as the harmonic mean of the average accuracy on known classes (OS*) and the accuracy on unknown classes (Unk). This metric addresses the shortcomings of overall mean accuracy (OS), which can overestimate performance in open-set scenarios by neglecting unknown categories. For instance, a classifier that perfectly predicts all known classes (OS* = 100%) but completely fails on unknown classes (Unk = 0%) would still yield a deceptively high OS (e.g., 90.9% if there are ten known classes). As the number of known classes increases, OS becomes even less reflective of actual open-set performance. Thus, HOS provides a balanced evaluation by penalizing models that do not handle unknown classes. The HOS is calculated as

$$\text{HOS} = \frac{2 \times \text{OS}^* \times \text{Unk}}{\text{OS}^* + \text{Unk}}. \tag{16}$$

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TABLE I
CLASSIFICATION RESULTS ON THE PU–PC TASK

Class	Closed-set			Open-set								
	CDAN	CLDA	NOUN	UADAL	OSBP	DAMC	ANNA	STA	MTS	OMEGA	PUMCL	FedAdapt
1	98.80	96.72	95.60	95.53	5.75	**99.89**	29.54	99.77	98.27	2.71	99.52	95.09
2	92.14	88.75	91.16	91.32	83.29	91.60	**94.96**	77.70	72.76	88.28	83.41	40.35
3	56.94	82.04	42.64	35.53	4.20	69.05	3.31	47.67	52.66	**80.52**	26.85	73.89
4	10.01	24.38	2.49	16.08	1.09	16.53	9.31	81.98	10.19	4.44	8.19	**93.34**
5	99.89	100	97.34	100	99.79	100	99.61	100	99.96	94.51	100	**100**
6	60.32	36.21	48.25	**87.96**	82.55	18.73	15.04	39.64	48.73	68.12	2.49	54.14
7	73.66	40.35	60.31	1.71	11.92	16.66	**72.78**	58.17	60.16	40.41	13.32	43.12
Unk (%)	0	0	0	65.09	59.08	36.90	61.47	18.72	18.63	73.35	**95.35**	91.10
OS* (%)	70.25	66.92	62.54	61.16	41.23	58.92	46.36	**72.13**	63.25	54.14	47.68	71.42
HOS (%)	0	0	0	63.06	48.57	45.38	52.86	29.73	28.79	62.30	63.57	**80.06**
Kappa	28.07	26.58	25.82	49.95	25.38	36.01	40.75	37.80	30.48	42.84	59.25	**71.72**

TABLE II
CLASSIFICATION RESULTS ON THE HU13–HU18 TASK

Class	Closed-set			Open-set								
	CDAN	CLDA	NOUN	UADAL	OSBP	DAMC	ANNA	STA	MTS	OMEGA	PUMCL	FedAdapt
1	42.43	90.32	93.87	88.07	91.56	94.10	5.20	**99.55**	98.78	23.09	16.83	59.71
2	49.48	60.38	56.42	37.20	49.31	39.97	14.26	40.04	43.34	4.28	54.28	**87.59**
3	68.93	69.53	62.63	62.73	38.18	**70.59**	37.29	62.27	45.63	52.37	50.50	51.37
4	35.33	68.42	95.86	87.59	94.73	72.18	30.82	100	97.36	42.48	98.49	**100**
5	88.96	92.00	82.63	87.18	70.83	**91.21**	9.15	82.19	87.80	88.49	56.44	72.01
6	73.65	44.61	68.05	65.43	57.12	51.80	36.07	**77.63**	64.86	43.46	38.34	67.01
7	35.91	82.33	59.43	56.68	37.20	22.06	26.23	18.00	26.76	**80.70**	19.22	0.88
Unk (%)	0	0	0	22.28	42.25	29.01	68.64	15.50	16.36	14.01	**80.53**	66.56
OS* (%)	56.38	72.51	74.13	**69.27**	62.70	63.13	22.71	68.52	66.36	47.84	47.73	62.65
HOS (%)	0	0	0	33.72	50.48	39.76	34.14	25.29	26.26	21.66	59.94	**64.55**
Kappa	33.61	31.78	39.10	38.55	36.22	33.13	17.47	35.44	31.58	26.38	32.35	**47.65**

A high HOS indicates strong performance on both known and unknown categories, and this metric is not affected by the number of known classes.

2) Implementation Details: In each experiment, 180 labeled samples per class are randomly selected from the source domain to construct the training set. Ten clients are used in total, comprising five labeled source clients and five unlabeled target clients. The patch size for data samples is fixed at 7×7. Training is performed in two stages, with 20 epochs in the first stage and 300 epochs in the second stage, using a batch size of 32. Federated aggregation is performed per epoch, i.e., the server conducts FedAvg at the end of each epoch and broadcasts the updated global model to all clients. Network optimization is conducted using stochastic gradient descent (SGD) with a weight decay of 5×10^{-4} and a momentum of 0.9. The learning rates for the feature extractor E and the prototype-mapping head A are set to 0.001, while the boundary module R uses a learning rate of 0.0001. The learning rate for the domain discriminator D is dataset-dependent: it is set to 0.01 for PU–PC, and 0.0001 for HU13–HU18 and Dioni–Loukia, to accommodate different domain gaps. The proposed method is implemented using PyTorch 1.12.1 and executed on a GPU with CUDA 12.9.

C. Compared Methods

To assess the effectiveness of the proposed approach, we conduct comparative experiments against three representative closed-set UDA baselines (CDAN [58], CLDA [59], and NOUN [60]) and eight recent OSDA methods (UADAL [56], OSBP [20], DAMC [55], ANNA [61], STA [21], MTS [22], OMEGA [23], and PUMCL [24]). Detailed descriptions and key differences are provided in the Supplementary Material Section I.

D. Experiments

1) PU–PC: The quantitative comparisons on the PU–PC task are reported in Table I. Overall, the proposed method demonstrates superior performance relative to existing approaches. Although several competing methods yield competitive accuracy on known classes, their ability to distinguish unknown classes remains limited. PUMCL performs well in detecting unknown samples, reaching an accuracy of 99.88%, yet suffers from reduced classification performance on known categories. By contrast, the proposed method achieves a more robust tradeoff between known and unknown class recognition, with OS*, Unk, and HOS values of 71.42%, 91.10%, and 80.06%, respectively. This results in an HOS improvement of 17.00% over the strongest baseline. The corresponding classification maps are presented in Fig. 3.

A t-SNE visualization on the representative PU–PC task is further provided in the Supplementary Material Section III, which gives an intuitive illustration of known-class alignment and unknown-sample separation.

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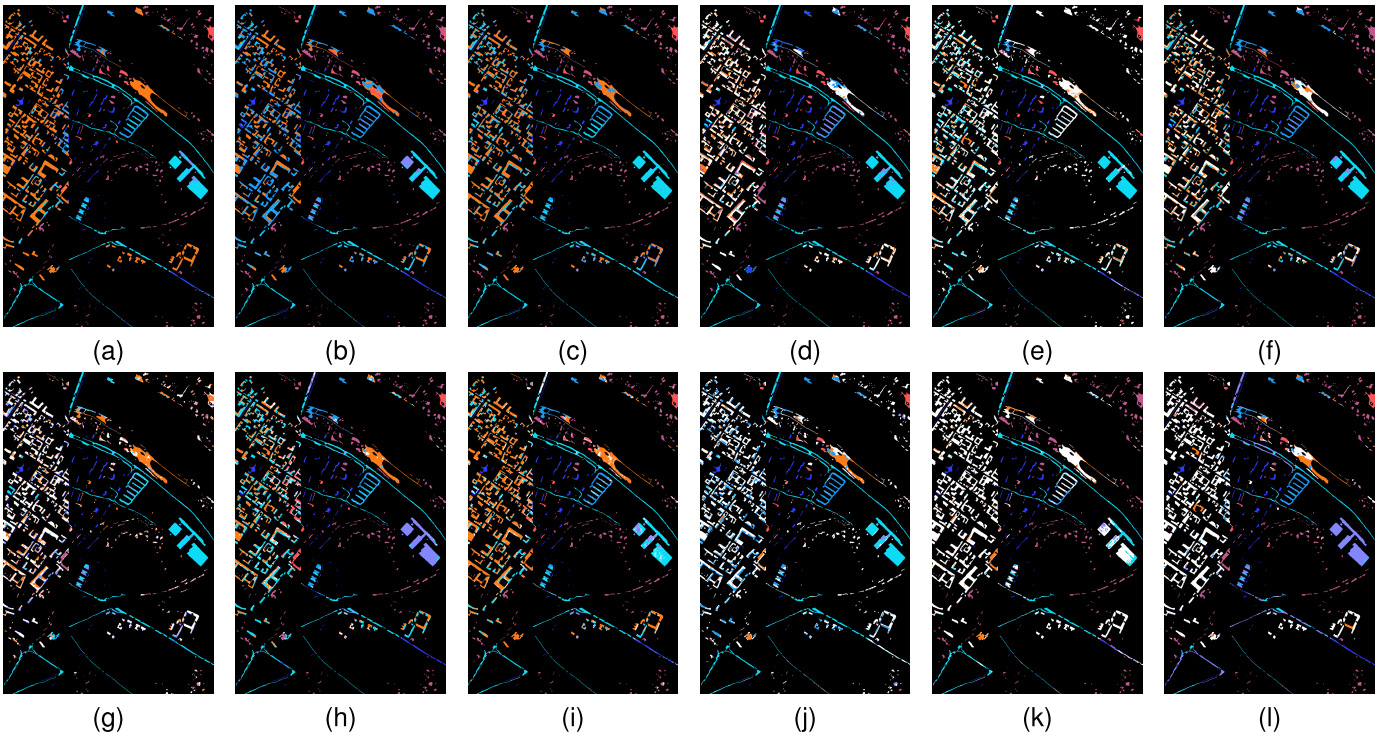


Fig. 3. Classification maps of the PU–PC task. (a) CDAN. (b) CLDA. (c) NOUN. (d) UADAL. (e) OSBP. (f) DAMC. (g) ANNA. (h) STA. (i) MTS. (j) OMEGA. (k) PUMCL. (l) FedAdapt.

TABLE III
CLASSIFICATION RESULTS ON THE DIONI–LOUKIA TASK

Class	Closed-set			Open-set								
	CDAN	CLDA	NOUN	UADAL	OSBP	DAMC	ANNA	STA	MTS	OMEGA	PUMCL	FedAdapt
1	8.73	0.97	6.31	2.91	7.76	2.42	3.39	**32.52**	4.85	13.10	25.24	13.11
2	100	87.03	90.74	79.62	50.01	100	70.37	7.40	46.29	96.29	81.48	**100**
3	95.07	59.15	45.30	56.33	29.81	63.14	**81.69**	41.78	81.22	70.18	20.89	54.23
4	73.41	100	94.93	**100**	15.18	92.40	97.46	54.43	79.74	59.49	0	58.23
5	4.78	0	1.98	0.09	0.09	0.09	7.67	1.17	0.18	16.98	0.99	**46.88**
6	64.69	86.01	82.93	83.64	57.34	74.88	53.08	49.76	66.82	**91.94**	44.54	27.73
7	39.60	44.04	43.58	9.95	10.16	1.22	34.73	12.32	53.74	23.63	**89.87**	5.63
8	18.10	70.19	59.16	74.17	22.95	65.78	9.71	77.26	**78.58**	4.63	0.44	63.79
9	92.63	99.52	99.04	99.52	98.81	0.71	90.97	**100**	99.04	96.67	97.62	86.69
Unk (%)	0	0	0	10.96	54.51	45.30	6.95	50.81	1.25	11.27	14.11	**57.04**
OS* (%)	55.22	60.77	58.22	56.25	32.46	44.51	49.90	41.85	**56.72**	52.55	40.12	50.69
HOS (%)	0	0	0	18.35	40.69	44.90	12.20	45.90	2.45	18.57	20.88	**53.68**
Kappa	16.64	18.07	16.82	12.72	7.89	12.80	15.53	19.01	19.39	13.68	12.08	**25.69**

2) Hu13–Hu18: Table II reports the quantitative comparison results of different methods on the HU13–HU18 task. Overall, the proposed approach demonstrates superior performance compared with the competing methods. UADAL achieves the highest accuracy on known-class classification, with OS* reaching 69.27%; however, its capability to recognize unknown classes is limited, as indicated by an Unk score of only 22.28%, which consequently leads to a low HOS value. In contrast, ANNA excels in identifying unknown classes, attaining an Unk accuracy of 68.64%, but exhibits relatively poor performance on known classes, with an OS* of 22.71% and an HOS of 34.14%. The proposed method achieves a more balanced performance across both known and unknown categories, yielding OS* and Unk values of 62.65% and 66.56%, respectively, and an overall HOS of 64.55%. Notably, the HOS is improved by 5.73% compared with the best-performing competing method. The corresponding classification maps for the proposed and comparison methods are illustrated in Fig. 4.

3) Dioni–Loukia: Table III presents the comparative results obtained on the Dioni–Loukia task. From the perspective of unknown-class recognition, the proposed method achieves the most competitive performance among all evaluated approaches. Specifically, it obtains Unk scores of 57.04%, respectively, resulting in an HOS value of 53.68%. This represents an improvement of 8.78% in HOS over the strongest competing method. Fig. 5 illustrates the classification maps generated by the proposed method in comparison with the other approaches.

4) ZY–GF: As reported in Table SIX of the Supplementary Material, the proposed method achieves the best HOS of 89.97%, outperforming the second-best method OMEGA by

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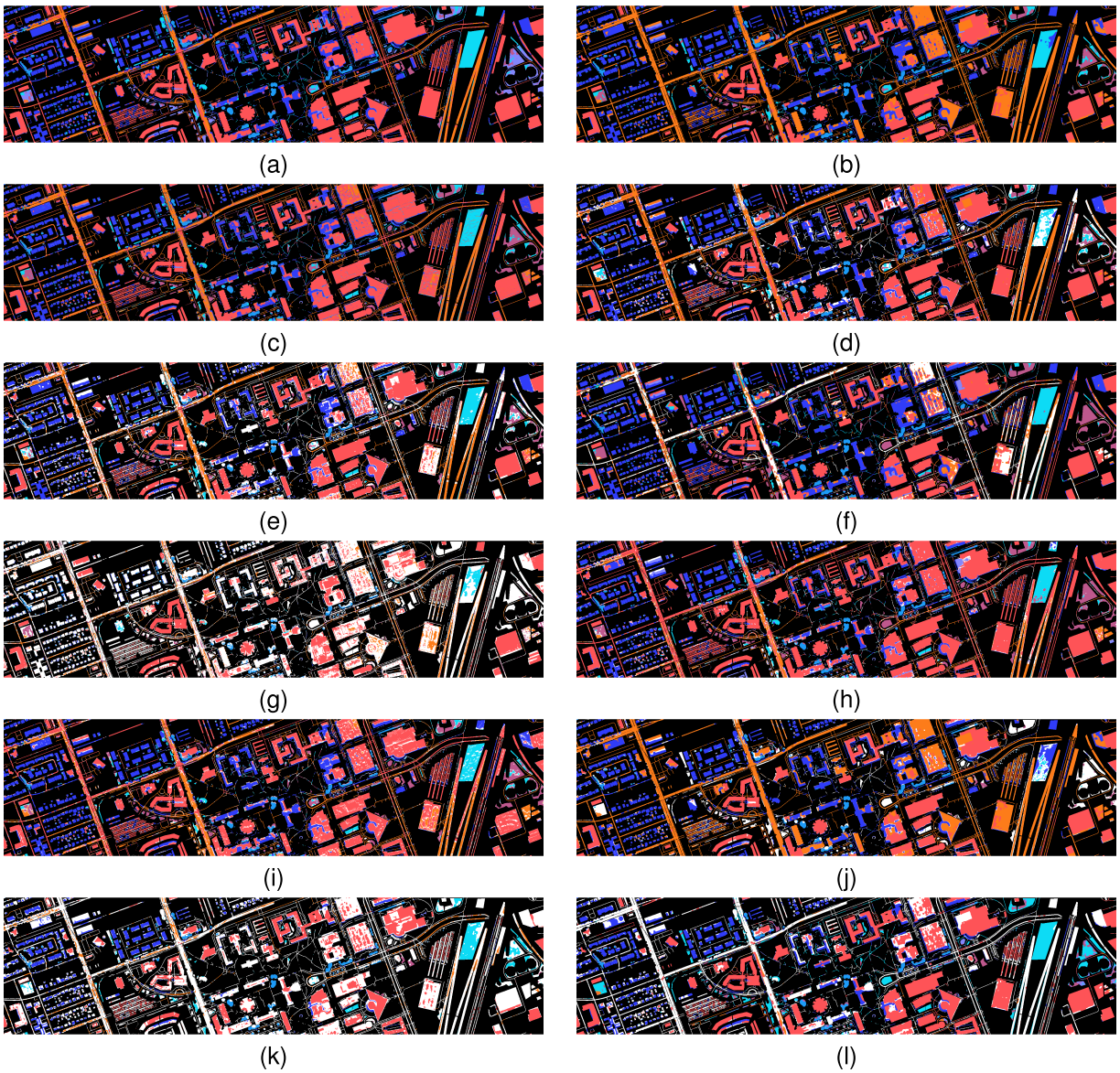


Fig. 4. Classification maps of the HU13–HU18 task. (a) CDAN. (b) CLDA. (c) NOUN. (d) UADAL. (e) OSBP. (f) DAMC. (g) ANNA. (h) STA. (i) MTS. (j) OMEGA. (k) PUMCL. (l) FedAdapt.

16.58%. Although OMEGA and ANNA obtain high Unk accuracies, their OS* values are much lower, indicating limited recognition of known classes. In contrast, our method maintains high OS* and Unk accuracies of 86.23% and 94.06%, respectively, showing a better balance between known-class adaptation and unknown-class rejection. The corresponding classification maps are shown in Fig. S6 in the Supplementary Material.

E. Ablation Studies

To verify the contribution of each component in FedAdapt, we conduct ablation studies on four cross-scene tasks. Specifically, Source-only denotes the model trained only with labeled source-domain samples, without target-domain adaptation. Without DA removes the adversarial domain alignment module, while keeping the prototype-based learning and dynamic thresholding mechanism. Without DT removes the dynamic threshold learning module and uses a fixed threshold for open-set inference. CPV + DA + DT is the basic variant that uses CPV, standard domain alignment, and standard dynamic thresholding. CPV + DA + WDT introduces weighted dynamic thresholding, and CPV + WDA + DT introduces weighted domain alignment. FedAdapt is the complete model with both weighted domain alignment and weighted dynamic thresholding.

As shown in Table IV, source-only obtains the lowest HOS on most tasks, indicating that source-domain training alone is insufficient under cross-satellite domain shift. Removing DA causes clear performance degradation compared with FedAdapt, which confirms that domain alignment is important for reducing the distribution discrepancy between source and target domains. Removing DT also decreases HOS, especially on the HU13–HU18 task, showing that a fixed threshold is less

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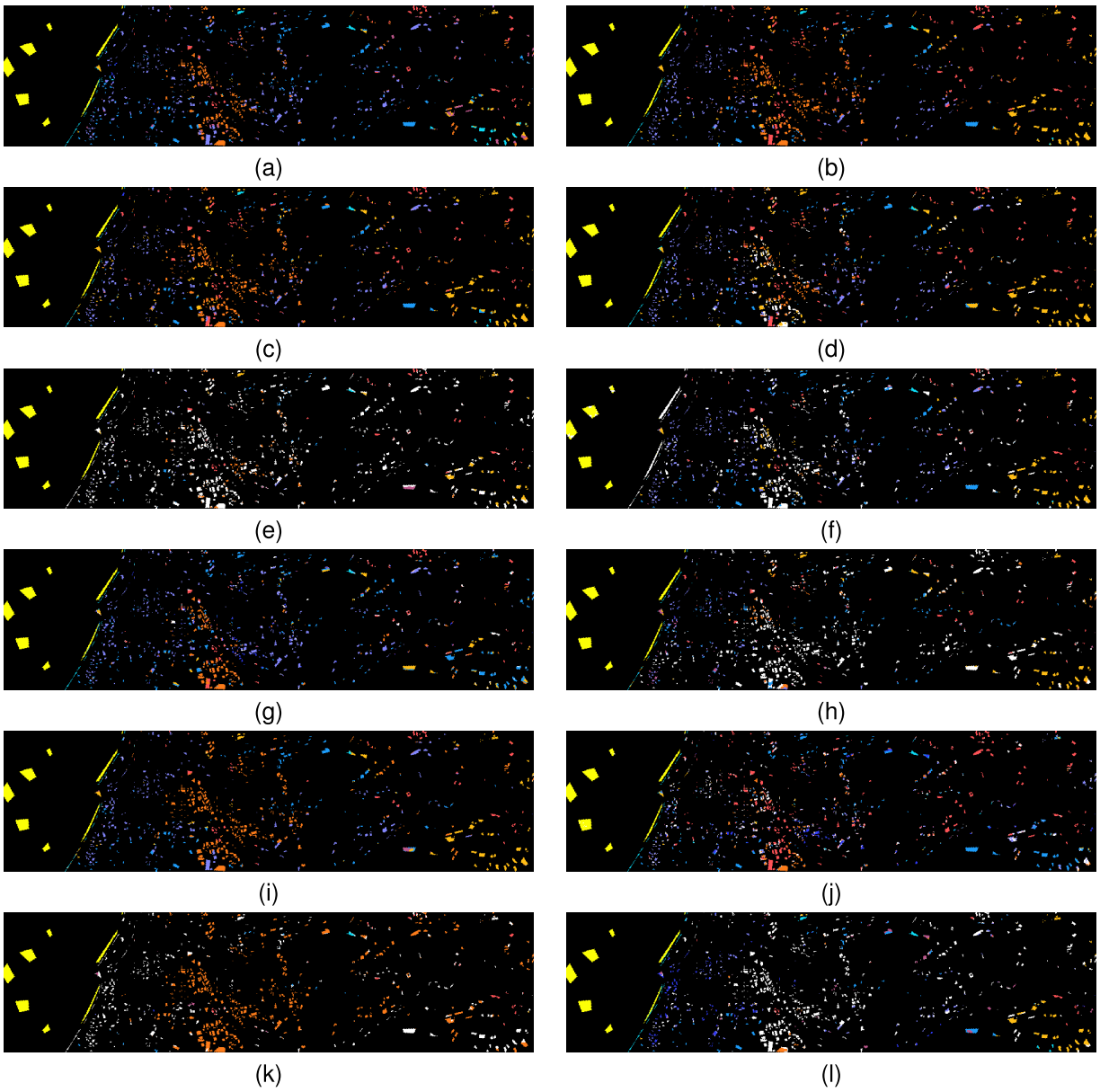


Fig. 5. Classification maps of the Dioni–Loukia task. (a) CDAN. (b) CLDA. (c) NOUN. (d) UADAL. (e) OSBP. (f) DAMC. (g) ANNA. (h) STA. (i) MTS. (j) OMEGA. (k) PUMCL. (l) FedAdapt.

TABLE IV
RESULTS ON THE PU–PC, HU13–HU18, DIONI–LOUKIA, AND ZY–GF TASKS IN THE ABLATION STUDY

Method	PU–PC	HU13–HU18	Dioni–Loukia	ZY–GF
Source-only	36.22	24.32	39.02	20.50
w/o DA	52.63	43.02	42.57	63.27
w/o DT	57.02	31.45	38.89	47.60
CPV + DA + DT	57.84	52.42	45.61	50.17
CPV + DA + WDT	64.53	57.94	51.27	64.45
CPV + WDA + DT	72.91	61.06	44.50	71.07
FedAdapt	**80.06**	**64.55**	**53.68**	**89.97**

reliable when the target domain contains unknown categories and large domain variations.

Compared with CPV + DA + DT, CPV + DA + WDT improves HOS from 57.84%, 52.42%, 45.61%, and 50.17% to 64.53%, 57.94%, 51.27%, and 64.45% on the four tasks, respectively. This demonstrates that the reliability-guided weighting strategy helps the threshold learning process by reducing the influence of unreliable target samples. In addition, CPV + WDA + DT improves the results on most tasks, and the complete FedAdapt further achieves the best performance on all tasks, with 80.06%, 64.55%, 53.68%, and 89.97%, respectively. The effectiveness of each component of the proposed method is demonstrated through ablation studies.

An additional analysis of negative transfer suppression is provided in the Supplementary Material Section IV. By measuring the known-class accuracy drop after introducing unknown target samples, the results show that relevance-driven weighting reduces the influence of potential unknown samples during alignment.

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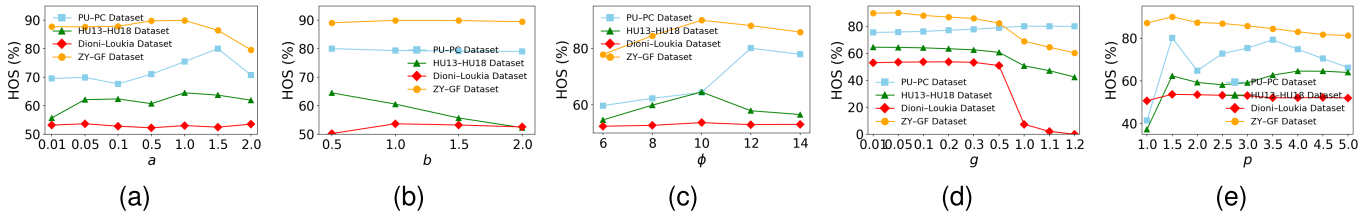


Fig. 6. Parameter analysis. The optimal values of (a) a, (b) b, (c) ϕ, (d) g, and (e) p are selected via grid search.

F. Parameter Analysis

We further investigate the sensitivity of the proposed model to its hyperparameters by employing a grid search strategy. Concretely, multiple alternative values are prescribed for a, b, ϕ, g, and p. The model is trained with every candidate setting, and the resulting test performance is collected. The optimal set of hyperparameters is then selected by comparing these outcomes.

1) We vary the hyperparameter a over the set {0.01, 0.05, 0.1, 0.5, 1.0, 1.5, 2.0}, while keeping the remaining hyperparameters fixed. As illustrated in Fig. 6(a), the HOS score first decreases and then increases as a becomes larger for the PU–PC dataset, the best HOS is obtained at $a = 1.5$, whereas for the HU13–HU18 dataset, the peak performance appears at $a = 1.0$. On the Dioni–Loukia dataset, the variation of HOS with respect to a is relatively small, and the maximum value is also achieved when $a = 0.05$. For the ZY–GF dataset, HOS first increases and then decreases as a increases, with the best performance obtained at $a = 1.0$.

2) The hyperparameter b is varied in the set {0.5, 1.0, 1.5, 2.0}, while all other hyperparameters are kept fixed. As shown in Fig. 6(b), on the PU–PC and HU13–HU18 datasets, the HOS tends to saturate, their maximum are at $b = 0.5$. For the Dioni–Loukia, HOS exhibits an initial increase followed by a subsequent decline, with the best performance obtained at $b = 1.0$. On the ZY–GF dataset, the HOS values remain relatively stable under different b values, and the highest HOS is achieved when $b = 1.0$.

3) The hyperparameter ϕ is varied in the set {6, 8, 10, 12, 14}, while the remaining hyperparameters are kept fixed. As illustrated in Fig. 6(c), on the PU–PC and HU13–HU18 datasets, HOS first increases markedly and then decreases, with the best performance obtained at $\phi = 12$ and $\phi = 10$. For the Dioni–Loukia dataset, the HOS score changes only slightly as ϕ increases, and its best value is obtained at $\phi = 10$. For the ZY–GF dataset, HOS increases clearly when ϕ grows from 6 to 10, and then gradually decreases when ϕ becomes larger. The best result is obtained at $\phi = 10$.

4) The hyperparameter g is varied in the set {0.01, 0.05, 0.1, 0.2, 0.3, 0.5, 1.0, 1.1, 1.2}, while the other hyperparameters are kept fixed. As shown in Fig. 6(d), for the HU13–HU18 and Dioni–Loukia datasets, HOS remains relatively flat initially and then drops sharply, with the best performance obtained at $g = 0.01$ and $g = 0.2$. While on the PU–PC dataset, the HOS score changes only slightly as g increases, and its best value is obtained at $g = 1.1$. On the ZY–GF dataset, HOS achieves its highest value at $g = 0.05$ and then decreases as g further increases.

5) The hyperparameter p is varied in the set {1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0}, while the other hyperparameters are kept fixed. As shown in Fig. 6(e), the HOS score increases sharply at first, and then levels off on the PU–PC and HU13–HU18 datasets, and the best values are obtained at $p = 1.5$ and $p = 4.0$. While on the Dioni–Loukia dataset, the HOS score changes only slightly as p increases, and its best value is obtained at $p = 1.5$. For the ZY–GF dataset, the best HOS is obtained when $p = 1.5$, after which the performance gradually decreases with larger p values.

Based on the above analysis, the proposed model can not only classify target-domain samples from known classes with high accuracy, but also effectively detect samples from unknown classes. Consequently, on the PU–PC dataset, the hyperparameters a, b, ϕ, g, and p are set to 1.5, 0.5, 12, 1.1, and 1.5, respectively. For the HU13–HU18 dataset, these hyperparameters are configured as 1.0, 0.5, 10, 0.01, and 4, respectively. On the Dioni–Loukia dataset, the settings are 0.05, 1.0, 10, 0.2, and 1.5, respectively. For the ZY–GF dataset, the corresponding settings are 1.0, 1.0, 10, 0.05, and 1.5, respectively.

Although the optimal hyperparameter values are not identical across different tasks, this variation is expected because the four adaptation scenarios exhibit different domain gaps, spectral characteristics, and openness levels. More importantly, the sensitivity curves in Fig. 6 show that FedAdapt does not depend on a single isolated parameter setting. For most hyperparameters, the performance remains relatively stable within a reasonable range, while clear degradation mainly appears when the parameters are set to overly small or overly large values. In practice, we adopt a lightweight coarse-search strategy to select the hyperparameters once for each adaptation task, and the selected setting is then shared by all clients in the federated training process. Therefore, no client-specific parameter tuning is required. In addition, the rejection boundary for unknown samples is learned through the proposed energy-based dynamic thresholding module, which avoids manually fixing an unknown-class threshold during testing.

To further examine the deployment feasibility of FedAdapt, detailed computational and communication profiling results are provided in the Supplementary Material Section V.

V. Conclusion

This article addresses the challenges of multisatellite HSI classification under limited downlink capacity, scarce annotations, substantial domain shift, and the presence of

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target-private unknown classes. FedAdapt, an open-set unsupervised federated domain adaptation framework, is proposed, which enables privacy-preserving on-orbit collaborative learning without transmitting raw HSI data to ground stations. FedAdapt constructs a prototype-induced metric space to stabilize the geometry of known classes, leverages relevance-driven sample weighting to perform cautious domain alignment and mitigate negative transfer, and learns an energy-based dynamic rejection boundary to separate known and unknown target samples without relying on a fixed threshold. Extensive cross-scene experiments demonstrate consistent improvements in open-set performance, validating the effectiveness and robustness of the proposed method. Future work will explore mechanisms for stronger privacy preservation and robustness under more complex open-set and non-IID conditions, thereby further enhancing the applicability of the proposed method.

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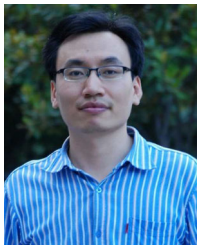
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